

# Bounded Staleness for Edge Caches

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Proceedings of the Example Conference (EXC '24), Lisbon, May 20–23, 2024, pp. 1–12 · doi:10.5555/exc24.0001

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**ABSTRACT** Edge caches answer reads close to users, but a cache that misses an invalidation can serve stale data for as long as it likes. We bound that staleness. Each replica holds a lease that expires before it can fall more than a configured bound behind, and a replica whose lease has lapsed sends reads to the origin until it catches up. On a synthetic edge workload across three regions, this lowers p99 read latency by up to 40% compared with strong reads, while no read is more than 250 ms stale.

## 1 INTRODUCTION

Caching at the edge trades freshness for latency, and most deployments leave the trade implicit: a value stays until an invalidation arrives, and invalidations can be delayed or lost. Applications that need fresh data then bypass the cache and pay the full round trip on every read.

We ask for less than strong consistency and more than none: a bound on how stale any read can be, which the operator sets per key space. Section 2 describes the lease protocol that enforces it, and Section 3 measures what it costs.

## 2 DESIGN

The origin grants each replica a lease of length  $L$  and promises not to acknowledge a write until every replica holding a lease has seen it or the lease has expired. A replica serves reads locally only while its lease is valid, so with clock skew at most  $d$ , no read is older than  $L + d$ . Renewals ride on the invalidation stream, so a healthy replica never pauses.

## 3 EVALUATION

We replayed a synthetic read-heavy workload against three regions with the bound set to 250 ms.

| Configuration         | p50 | p99 | Max staleness |
|-----------------------|-----|-----|---------------|
| Strong reads (origin) | 21  | 48  | 0 ms          |
| Invalidation only     | 3   | 18  | unbounded     |
| Bounded staleness     | 4   | 29  | 250 ms        |

Table 1. Read latency in milliseconds (lower is better).

## 4 CONCLUSION

A lease per replica turns "eventually fresh" into "fresh within a bound the operator chose", at a fraction of the latency of strong reads. The code and traces are released with the paper.

## REFERENCES

1. A. Author and P. Investigator. Leases for replicated caches. *Journal of Example Systems*, 2021.
2. P. Investigator. Measuring staleness in the wild. In *Proc. EXC '22*, 2022.